

- 1) (b) Data and a pointer to the next node.
- 2) (c) NULL
- 3) (b) Self-referential structure
- 4) (b) `head == NULL`
- 5) (b) `newNode → next = head; head = newNode;`
- 6) (b) `temp = head; head = head → next, free(temp);`
- 7) (c) Traversing from tail to head.
- 8) (b) `n`
- 9) (c) copying the data of `p → next` into `p`, then deleting `p → next`.
- 10) (b) `while (p != NULL)`
- 11) (b) `p → next == NULL`
- 12) (c) Two pointers advancing at different rate.
- 13) (c) Three
- 14) (b) Exactly one node.
- 15) (b) `P = (struct node *) malloc (sizeof (struct Node));`
- 16) (b) Avoid special-case handling for insertion deletion at the head.
- 17) (c) The first node.
- 18) (b) It does not support random (indexed) access.
- 19) (b) Loss of access to all ~~nodes~~ nodes, causing a memory leak.
- 20) (b) A slow pointer and a fast pointer.
- 21) (c) The last node.
- 22) (b) Nodes may be scattered anywhere in memory
- 23) `struct node {`
`int data;`
`struct node * next;`
`};`
- 24) The code prints the sum of each node and its next node :
`→ 10 + 20 = 30`
`→ 20 + 30 = 50`
`→ 30 + 40 = 70`
 • Output is → 30 50 70
- 25) Error:- head is passing by value, so changes

to the head pointer won't affect the original list. Also, an empty-list case needs handling.
correct version:-

```
void insertEnd(struct node **head, int n)
{
    struct node *t = (struct node *)malloc(
        sizeof(struct node));
    t->data = n;
    t->next = NULL;
    if (*head == NULL) {
        *head = t;
        return;
    }
    struct node *p = *head;
    while (p->next != NULL) {
        p = p->next;
    }
    p->next = t;
}
```

```
26) struct node *reverse(struct node *head) {
    struct node *prev = NULL;
    struct node *curr = head;
    struct node *next = NULL;
    while (curr != NULL) {
        next = curr->next;
        curr->next = prev;
        prev = curr;
        curr = next;
    }
    return prev;
}
```

27) Suppose, the list is $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \Rightarrow \text{NULL}$
 $\Rightarrow$ I want to delete 30.
 $\Rightarrow$ Then, first 20 nodes next pointer needs to be pointed to 40.

Before $\rightarrow$ ~~40~~ 10 $\rightarrow$ 20 $\rightarrow$ 30 $\rightarrow$ NULL
 prev[↑] del Next

after $\rightarrow$ 10 $\rightarrow$ 20 $\rightarrow$ 40 $\rightarrow$ NULL
 ↑

30 deleted.

prev $\rightarrow$ next = del $\rightarrow$ next;
 free (del);

28) Initial list :-

5 $\rightarrow$ 8 $\rightarrow$ 12 $\rightarrow$ 3 $\rightarrow$ NULL

(i) 7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 12 $\rightarrow$ 3 $\rightarrow$ NULL

(ii) 7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 3 $\rightarrow$ NULL

(iii) 7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 3 $\rightarrow$ 9 $\rightarrow$ NULL

Ans:- 7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 3 $\rightarrow$ 9 $\rightarrow$ NULL